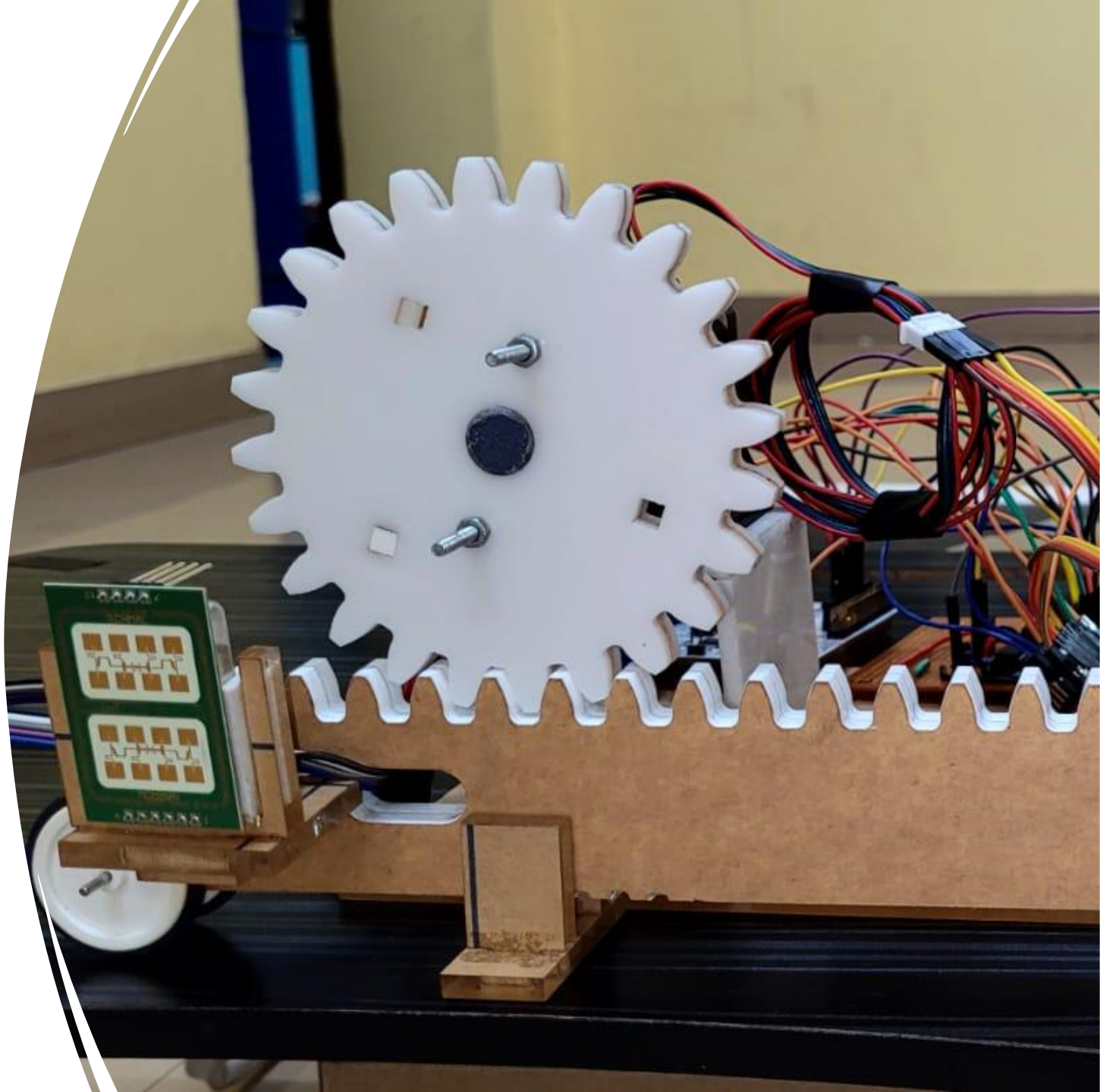
MON-12, Team SKORK

GROUND PENETRATING RADAR

VISION BEYOND BOUNDARIES

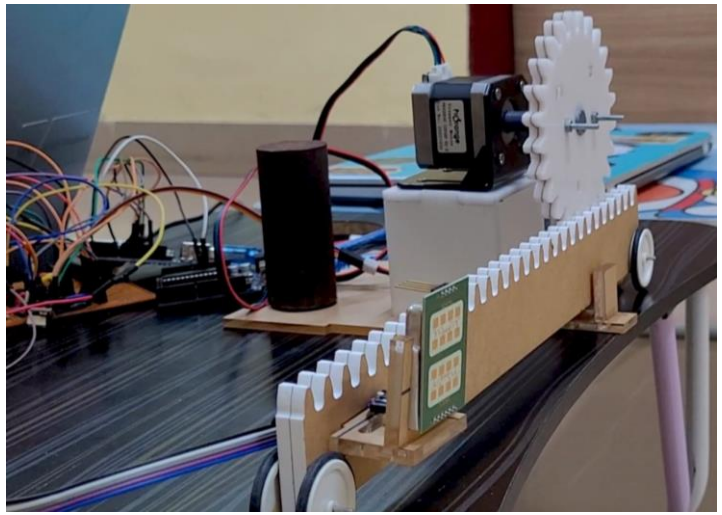
Our enhanced GPR detects underground utilities, hidden objects behind walls, and anomalies at a distance. It helps construction workers, engineers, and surveyors prevent accidental damage and ensure safe, informed excavation and site assessment.

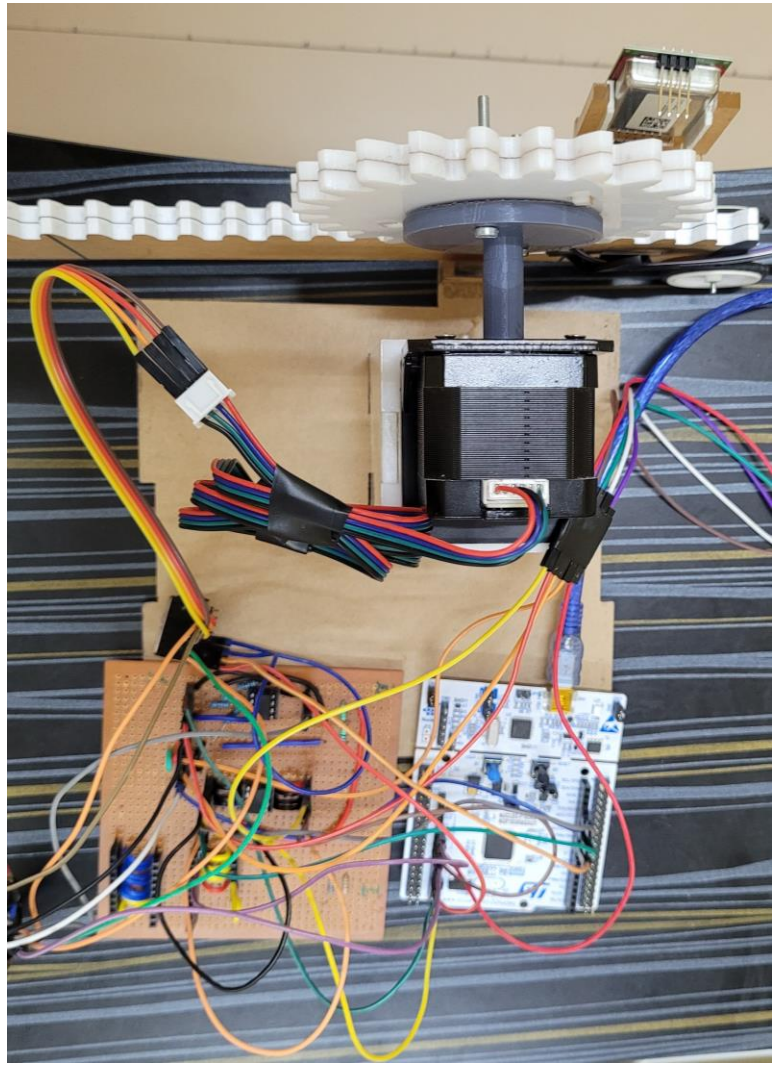


Status of Project

Key specifications and requirements proposed at beginning of semester	**Status at end of semester**	**Remarks/Justification**
Completely automated system (along with a GUI)	Partially complete	The step of signal processing using MATLAB still needs initiation, post data collection using python
Independent portable system	Partially complete	Power supply to stepper motor using 12V adapter, and STM32 connection with PC for serial connection is still there
Automated maneuver of FMCW module with precise displacement steps	Complete	

Photographs of Final Prototype





Final Demo Video

https://drive.google.com/file/d/1dM9vwxxUahr5V7ky8aHOwfB1LIrTCQWT/view?usp=sharing